

Experiment/ Case Study No.: 5

1. **Experiment Title:** Apply Array Methods and Object Handling in JavaScript

2. **Software/Tools Required:** VS Code, Web Browser, HTML, CSS, JavaScript

3. **Experiment Program Code:**

product.html

```
<!DOCTYPE html>
<html>

<head>
  <meta charset="UTF-8">
  <title>Product Calculation</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #e8f5e9;
      margin: 50px auto;
      max-width: 400px;
    }

    .container {
      background: #ffffff;
      padding: 20px;
      border: 2px solid #4caf50;
      border-radius: 8px;
    }

    h2 {
      text-align: center;
      color: #2e7d32;
    }

    .form-group {
      margin-bottom: 15px;
    }

    label {
      display: block;
      margin-bottom: 5px;
      font-weight: bold;
      color: #333;
    }

    input {
      width: 100%;
      padding: 8px;
      box-sizing: border-box;
      border: 1px solid #ccc;
      border-radius: 4px;
    }

    input:focus {
      border-color: #4caf50;
      outline: none;
    }

    button {
      width: 100%;
      padding: 10px;
      background-color: #4caf50;
      color: white;
      border: none;
      border-radius: 4px;
      font-size: 16px;
      cursor: pointer;
    }

    button:hover {
      background-color: #388e3c;
    }

    #result {
      margin-top: 20px;
      padding: 15px;
      background-color: #c8e6c9;
      border: 1px solid #4caf50;
      border-radius: 4px;
      display: none;
    }

    .result-item {
      display: flex;
      justify-content: space-between;
      margin-bottom: 5px;
    }
```

```

    }

    hr {
        border: 0;
        border-top: 1px solid #4caf50;
        margin: 10px 0;
    }
</style>
</head>

<body>
    <div class="container">
        <h2>Product Invoice</h2>
        <div class="form-group">
            <label for="customerName">Customer Name</label>
            <input type="text" id="customerName" placeholder="Enter customer name">
        </div>
        <div class="form-group">
            <label for="product">Product Name</label>
            <input type="text" id="product" placeholder="Enter product name">
        </div>
        <div class="form-group">
            <label for="price">Price</label>
            <input type="number" id="price" placeholder="Enter price" min="0">
        </div>
        <div class="form-group">
            <label for="quantity">Quantity</label>
            <input type="number" id="quantity" placeholder="Enter quantity" min="1" value="1">
        </div>
        <button onclick="calculateBill()">Calculate Bill</button>

        <div id="result">
            <div class="result-item"><span>Customer:</span> <strong id="res-customer"></strong></div>
            <div class="result-item"><span>Product:</span> <strong id="res-product"></strong></div>
            <div class="result-item"><span>Subtotal:</span> <strong id="res-subtotal"></strong></div>
            <div class="result-item"><span>GST (18%):</span> <strong id="res-gst"></strong></div>
            <hr>
            <div class="result-item"><span>Total:</span> <strong id="res-total"></strong></div>
        </div>
    </div>

    <p style="margin-top: 20px; color: #2e7d32; font-weight: bold;">Made by Ayush Bisen</p>
    <p style="text-align: center;"></p>

    <script src="script.js"></script>
</body>
</html>

```

script.js

```

function calculateBill() {
  // let example
  let customerName = document.getElementById("customerName").value;
  let product = document.getElementById("product").value;
  let price = Number(document.getElementById("price").value);
  let quantity = Number(document.getElementById("quantity").value);

  if (!customerName || !product || price <= 0 || quantity <= 0) {
    alert("Please enter valid details for all fields.");
    return;
  }

  const GST = 0.18;

  let subtotal = price * quantity;
  let gstAmount = subtotal * GST;
  let total = subtotal + gstAmount;

  const bill = {
    customerName,
    product,
    subtotal,
    gstAmount,
    total
  };

  document.getElementById("res-customer").textContent = bill.customerName;
  document.getElementById("res-product").textContent = bill.product;
  document.getElementById("res-subtotal").textContent = bill.subtotal.toFixed(2);
  document.getElementById("res-gst").textContent = bill.gstAmount.toFixed(2);
  document.getElementById("res-total").textContent = bill.total.toFixed(2);

  document.getElementById("result").style.display = "block";

  console.log("Calculation completed.");
  console.log("Bill generated for:", bill.customerName);
  console.log("Total Amount:", bill.total);
}

```

shopping.html

```

<!DOCTYPE html>
<html>

<head>
  <meta charset="UTF-8">
  <title>Online Shopping Bill Calculator</title>
  <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/font-awesome@6.4.0/css/all.min.css">
  <style>
    :root {
      --primary-green: #2e7d32;
      --light-green: #e8f5e9;
      --hover-green: #388e3c;
      --accent-green: #4caf50;
      --dark-green: #1b5e20;
      --bg-color: #f1f8e9;
      --text-color: #333;
    }

    body {
      font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
      background-color: var(--bg-color);
      margin: 0;
      padding: 20px;
      color: var(--text-color);
    }

    .container {
      max-width: 1200px;
      margin: 0 auto;
      background: #ffffff;
      padding: 30px;
      border-radius: 12px;
      box-shadow: 0 4px 15px rgba(46, 125, 50, 0.1);
      border-top: 5px solid var(--primary-green);
    }

    h1,
    h2 {
      text-align: center;
      color: var(--dark-green);
      margin-top: 0;
    }

    .input-section {
      display: grid;

```

```
    grid-template-columns: repeat(auto-fit, minmax(150px, 1fr));
    gap: 15px;
    margin-bottom: 20px;
}

.form-group {
    display: flex;
    flex-direction: column;
}

label {
    margin-bottom: 5px;
    font-weight: 600;
    color: var(--primary-green);
}

input {
    padding: 10px;
    border: 1px solid #c8e6c9;
    border-radius: 6px;
    font-size: 14px;
    transition: all 0.3s;
}

input:focus {
    border-color: var(--accent-green);
    box-shadow: 0 0 5px rgba(76, 175, 80, 0.3);
    outline: none;
}

input:disabled {
    background-color: #f5f5f5;
    cursor: not-allowed;
}

button {
    padding: 12px 20px;
    background-color: var(--accent-green);
    color: white;
    border: none;
    border-radius: 6px;
    font-size: 16px;
    font-weight: bold;
    cursor: pointer;
    transition: background-color 0.3s, transform 0.1s;
}

button:hover {
    background-color: var(--hover-green);
}

button:active {
    transform: scale(0.98);
}

button.secondary {
    background-color: var(--primary-green);
}

button.secondary:hover {
    background-color: var(--dark-green);
}

.actions {
    display: flex;
    gap: 10px;
    justify-content: center;
    margin-bottom: 30px;
}

table {
    width: 100%;
    border-collapse: collapse;
    margin-bottom: 20px;
}

th,
td {
    padding: 12px;
    text-align: left;
    border-bottom: 1px solid #c8e6c9;
}

th {
    background-color: var(--light-green);
    color: var(--dark-green);
}
```

```
    font-weight: 600;
  }

tr:hover {
  background-color: #f1f8e9;
}

.main-layout {
  display: flex;
  gap: 30px;
  align-items: flex-start;
}

.left-panel {
  flex: 1;
}

.right-panel {
  flex: 1;
  background: #f9fbf9;
  padding: 20px;
  border-radius: 8px;
  border: 1px solid #c8e6c9;
}

@media (max-width: 768px) {
  .main-layout {
    flex-direction: column;
  }
}

.product-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));
  gap: 20px;
  margin-bottom: 30px;
}

.product-card {
  background-color: white;
  border: 1px solid var(--accent-green);
  border-radius: 8px;
  padding: 15px;
  text-align: center;
  box-shadow: 0 2px 5px rgba(0, 0, 0, 0.05);
  transition: transform 0.2s;
}

.product-card:hover {
  transform: translateY(-3px);
  box-shadow: 0 4px 10px rgba(46, 125, 50, 0.15);
}

.product-card h3 {
  color: var(--primary-green);
  margin-top: 0;
  margin-bottom: 10px;
}

.product-card .price {
  font-size: 18px;
  font-weight: bold;
  color: var(--dark-green);
  margin: 10px 0 15px 0;
}

.product-card .add-section {
  display: flex;
  gap: 10px;
  justify-content: center;
}

.product-card input {
  width: 60px;
  padding: 8px;
  text-align: center;
}

.product-card button {
  padding: 8px 15px;
}

.invoice-section {
  background-color: var(--light-green);
  padding: 20px;
  border-radius: 8px;
}
```

```

        border: 1px dashed var(--accent-green);
        display: none;
    }

    .invoice-header {
        display: flex;
        justify-content: space-between;
        margin-bottom: 20px;
        border-bottom: 2px solid var(--primary-green);
        padding-bottom: 10px;
    }

    .summary-row {
        display: flex;
        justify-content: space-between;
        margin-bottom: 8px;
        font-size: 16px;
    }

    .summary-row.total {
        font-size: 20px;
        font-weight: bold;
        color: var(--dark-green);
        border-top: 2px solid var(--accent-green);
        padding-top: 10px;
        margin-top: 10px;
    }

    .delete-btn {
        background-color: #ef5350;
        padding: 6px 12px;
        font-size: 12px;
    }

    .delete-btn:hover {
        background-color: #d32f2f;
    }

    @media print {
        body {
            background-color: white;
            padding: 0;
        }

        .container {
            box-shadow: none;
            border: none;
            padding: 0;
        }

        .no-print {
            display: none !important;
        }

        .invoice-section {
            display: block !important;
            border: none;
            background-color: transparent;
            padding: 0;
        }

        .footer {
            text-align: center;
            margin-top: 20px;
            color: var(--primary-green);
            font-weight: bold;
        }

        #emptyCartMessage {
            text-align: center;
            color: #81c784;
            font-style: italic;
            padding: 20px;
        }
    }
</style>
</head>

<body>
    <div class="container">
        <h1 class="no-print"><i class="fas fa-leaf"></i> Checkout</h1>

        <div class="main-layout">
            <div class="left-panel no-print" id="shopContainer">
                <h2><i class="fas fa-store"></i> Shop Products</h2>
                <div class="product-grid">

```

```

<div class="product-card">
  <h3><i class="fas fa-solar-panel"></i> Solar Panels</h3>
  <p class="price">₹15000.00</p>
  <div class="add-section">
    <input type="number" id="qty-solar" value="1" min="1">
    <button onclick="addToCart('Solar Panels', 15000, 'qty-solar')">Add</button>
  </div>
</div>
<div class="product-card">
  <h3><i class="fas fa-bolt"></i> Inverter</h3>
  <p class="price">₹10000.00</p>
  <div class="add-section">
    <input type="number" id="qty-inverter" value="1" min="1">
    <button onclick="addToCart('Inverter', 10000, 'qty-inverter')">Add</button>
  </div>
</div>
<div class="product-card">
  <h3><i class="fas fa-plug"></i> Power Cables</h3>
  <p class="price">₹500.00</p>
  <div class="add-section">
    <input type="number" id="qty-cables" value="1" min="1">
    <button onclick="addToCart('Power Cables', 500, 'qty-cables')">Add</button>
  </div>
</div>
<div class="product-card">
  <h3><i class="fas fa-battery-full"></i> Battery</h3>
  <p class="price">₹12000.00</p>
  <div class="add-section">
    <input type="number" id="qty-battery" value="1" min="1">
    <button onclick="addToCart('Battery', 12000, 'qty-battery')">Add</button>
  </div>
</div>
</div>
</div>

<div class="right-panel">
  <div class="no-print">
    <div class="form-group" style="margin-bottom: 20px;">
      <label for="customerName">Customer Name</label>
      <input type="text" id="customerName" placeholder="Enter customer name">
    </div>

    <div class="manual-add-section"
      style="background: #e8f5e9; padding: 15px; border-radius: 8px; margin-bottom: 20px; border: 1px dashed
#4caf50;">

      <h3 style="margin-top:0; margin-bottom:10px; color: var(--primary-green); font-size: 16px;"><i
class="fas fa-plus-circle"></i> Custom Item</h3>
      <div style="display:flex; gap:10px; align-items:flex-end;">
        <div class="form-group" style="flex:2;">
          <label for="product" style="font-size: 13px;">Product</label>
          <input type="text" id="product" placeholder="Item name" style="padding: 8px;">
        </div>
        <div class="form-group" style="flex:1.5;">
          <label for="price" style="font-size: 13px;">Price (₹)</label>
          <input type="number" id="price" placeholder="0" min="0" step="0.01"
style="padding: 8px;">
        </div>
        <div class="form-group" style="flex:1;">
          <label for="quantity" style="font-size: 13px;">Qty</label>
          <input type="number" id="quantity" placeholder="1" min="1" value="1"
style="padding: 8px;">
        </div>
        <button onclick="addItem()"
style="padding: 9px 12px; background-color: var(--primary-green); margin-bottom: 2px;"><i
class="fas fa-cart-plus"></i></button>
      </div>
    </div>
  </div>

  <div id="cartSection" class="no-print">
    <table id="cartTable">
      <thead>
        <tr>
          <th>Product</th>
          <th>Price</th>
          <th>Qty</th>
          <th>Total</th>
          <th>Action</th>
        </tr>
      </thead>
      <tbody id="cartBody">
      </tbody>
    </table>
    <div id="emptyCartMessage">Your cart is currently empty.</div>

    <div class="actions" id="mainActions" style="margin-top: 20px;">

```

```

        <button class="secondary" onclick="generateInvoice()"><i class="fas fa-file-invoice"></i>
            Generate Invoice</button>
        <button class="secondary" onclick="showExpensiveItems()"><i class="fas fa-filter"></i>
            Expensive Items (>20k)</button>
    </div>
</div>

<div id="invoiceSection" class="invoice-section">
    <div class="invoice-header">
        <div>
            <h2 style="margin: 0; color: var(--dark-green);"><i class="fas fa-leaf"></i> GreenMart
                Invoice</h2>
            <p style="margin: 5px 0 0 0;" id="invDate"></p>
        </div>
        <div style="text-align: right;">
            <strong>Customer:</strong> <span id="invCustomer"></span><br>
            <strong>Invoice #:</strong> <span id="invNumber"></span>
        </div>
    </div>

    <table id="invoiceTable">
        <thead>
            <tr>
                <th>Product</th>
                <th>Price</th>
                <th>Qty</th>
                <th>Total</th>
            </tr>
        </thead>
        <tbody id="invoiceBody">
            <!-- Invoice items -->
        </tbody>
    </table>

    <div style="width: 100%; max-width: 300px; margin-left: auto; margin-top: 20px;">
        <div class="summary-row">
            <span>Total Items:</span>
            <strong id="res-items">0</strong>
        </div>
        <div class="summary-row">
            <span>Subtotal:</span>
            <strong id="res-subtotal">₹0.00</strong>
        </div>
        <div class="summary-row" id="discount-row" style="color: #388e3c; display: none;">
            <span>Discount:</span>
            <strong id="res-discount">-₹0.00</strong>
        </div>
        <div class="summary-row">
            <span>Tax (18% GST):</span>
            <strong id="res-gst">₹0.00</strong>
        </div>
        <div class="summary-row total">
            <span>Grand Total:</span>
            <strong id="res-total">₹0.00</strong>
        </div>
    </div>

    <div class="actions no-print" style="margin-top: 30px;">
        <button onclick="window.print()"><i class="fas fa-print"></i> Print Invoice</button>
        <button class="secondary" onclick="resetForm()"><i class="fas fa-plus-circle"></i> New
            Transaction</button>
    </div>
</div>
</div>
</div>

<p class="footer no-print">Made by Ayush Bisen</p>
</div>

<script src="shopping.js"></script>
</body>
</html>

```

4. Output:

alt text

5. Case Study Title: Cart total calculator with discount logic.

6. Case Study Program Code:

shopping.js

```

let cart = [];
const GST_RATE = 0.18;

document.addEventListener('DOMContentLoaded', () => {

```

```

    updateCartTable();
});

function addItem() {
    const customerNameInput = document.getElementById("customerName");
    const productInput = document.getElementById("product");
    const priceInput = document.getElementById("price");
    const quantityInput = document.getElementById("quantity");

    const customerName = customerNameInput.value.trim();
    const product = productInput.value.trim();
    const price = parseFloat(priceInput.value);
    const quantity = parseInt(quantityInput.value);

    if (!customerName) {
        alert("Please enter customer name.");
        customerNameInput.focus();
        return;
    }

    if (!product) {
        alert("Please enter product name.");
        productInput.focus();
        return;
    }

    if (isNaN(price) || price <= 0) {
        alert("Please enter a valid price.");
        priceInput.focus();
        return;
    }

    if (isNaN(quantity) || quantity <= 0) {
        alert("Please enter a valid quantity.");
        quantityInput.focus();
        return;
    }

    const itemTotal = price * quantity;

    const item = {
        id: Date.now(),
        product,
        price,
        quantity,
        total: itemTotal
    };

    cart.push(item);

    customerNameInput.disabled = true;

    // Clear product inputs
    productInput.value = "";
    priceInput.value = "";
    quantityInput.value = "1";
    productInput.focus();

    updateCartTable();
}

function addToCart(productName, price, qtyInputId) {
    const customerNameInput = document.getElementById("customerName");
    const quantityInput = document.getElementById(qtyInputId);

    const customerName = customerNameInput.value.trim();
    const quantity = parseInt(quantityInput.value);

    if (!customerName) {
        alert("Please enter customer name before adding items.");
        customerNameInput.focus();
        return;
    }

    if (isNaN(quantity) || quantity <= 0) {
        alert("Please enter a valid quantity.");
        quantityInput.focus();
        return;
    }

    const existingItem = cart.find(item => item.product === productName);
    if (existingItem) {
        existingItem.quantity += quantity;
        existingItem.total = existingItem.price * existingItem.quantity;
    } else {
        const itemTotal = price * quantity;

```

```

        const item = {
            id: Date.now() + Math.random(),
            product: productName,
            price,
            quantity,
            total: itemTotal
        };
        cart.push(item);
    }

    customerNameInput.disabled = true;
    quantityInput.value = "1";
    updateCartTable();
}

function removeItem(id) {
    cart = cart.filter(item => item.id !== id);
    updateCartTable();

    if (cart.length === 0) {
        document.getElementById("customerName").disabled = false;
        document.getElementById("invoiceSection").style.display = "none";
    }
}

function updateCartTable() {
    const cartBody = document.getElementById("cartBody");
    const cartTable = document.getElementById("cartTable");
    const emptyCartMessage = document.getElementById("emptyCartMessage");

    cartBody.innerHTML = "";

    if (cart.length === 0) {
        cartTable.style.display = "none";
        emptyCartMessage.style.display = "block";
    } else {
        cartTable.style.display = "table";
        emptyCartMessage.style.display = "none";

        cart.forEach(item => {
            const row = document.createElement("tr");
            row.innerHTML = `
                <td>${item.product}</td>
                <td>₹${item.price.toFixed(2)}</td>
                <td>${item.quantity}</td>
                <td>₹${item.total.toFixed(2)}</td>
                <td><button class="delete-btn" onclick="removeItem(${item.id})"><i class="fas fa-trash"></i></button></td>
            `;
            cartBody.appendChild(row);
        });
    }
}

function generateInvoice() {
    if (cart.length === 0) {
        alert("Cart is empty. Please add items to generate invoice.");
        return;
    }

    const customerName = document.getElementById("customerName").value.trim();

    // Populate invoice details
    document.getElementById("invCustomer").textContent = customerName;
    document.getElementById("invNumber").textContent = "INV-" + Math.floor(100000 + Math.random() * 900000);

    const options = { year: 'numeric', month: 'long', day: 'numeric', hour: '2-digit', minute: '2-digit' };
    document.getElementById("invDate").textContent = "Date: " + new Date().toLocaleDateString('en-IN', options);

    const invoiceBody = document.getElementById("invoiceBody");
    invoiceBody.innerHTML = "";

    let subtotal = 0;
    let totalItems = 0;

    cart.forEach(item => {
        subtotal += item.total;
        totalItems += item.quantity;

        const row = document.createElement("tr");
        row.innerHTML = `
            <td>${item.product}</td>
            <td>₹${item.price.toFixed(2)}</td>
            <td>${item.quantity}</td>
            <td>₹${item.total.toFixed(2)}</td>
        `;
        invoiceBody.appendChild(row);
    });
}

```

```

});

let discount = 0;
if (subtotal > 50000) {
    discount = subtotal * 0.15; // 15% discount
} else if (subtotal > 20000) {
    discount = subtotal * 0.10; // 10% discount
} else if (subtotal > 10000) {
    discount = subtotal * 0.05; // 5% discount
} else {
    discount = 0;
}

const discountedSubtotal = subtotal - discount;
const gstAmount = discountedSubtotal * GST_RATE;
const grandTotal = discountedSubtotal + gstAmount;

document.getElementById("res-items").textContent = totalItems;
document.getElementById("res-subtotal").textContent = "₹" + subtotal.toFixed(2);

const discountRow = document.getElementById("discount-row");
if (discount > 0 && discountRow) {
    discountRow.style.display = "flex";
    document.getElementById("res-discount").textContent = "-₹" + discount.toFixed(2);
} else if (discountRow) {
    discountRow.style.display = "none";
}

document.getElementById("res-gst").textContent = "₹" + gstAmount.toFixed(2);
document.getElementById("res-total").textContent = "₹" + grandTotal.toFixed(2);

document.getElementById("invoiceSection").style.display = "block";
document.getElementById("cartSection").style.display = "none";
document.getElementById("shopContainer").style.display = "none";
if (document.querySelector(".manual-add-section")) {
    document.querySelector(".manual-add-section").style.display = "none";
}
document.getElementById("mainActions").style.display = "none";
}

function resetForm() {
    cart = [];
    document.getElementById("customerName").value = "";
    document.getElementById("customerName").disabled = false;
    document.getElementById("product").value = "";
    document.getElementById("price").value = "";
    document.getElementById("quantity").value = "1";

    const qtyInputs = document.querySelectorAll('input[id^="qty-"]');
    qtyInputs.forEach(input => input.value = "1");

    document.getElementById("invoiceSection").style.display = "none";
    document.getElementById("cartSection").style.display = "block";
    document.getElementById("shopContainer").style.display = "block";
    if (document.querySelector(".manual-add-section")) {
        document.querySelector(".manual-add-section").style.display = "block";
    }
    document.getElementById("mainActions").style.display = "flex";

    updateCartTable();
}

function showExpensiveItems() {
    if (cart.length === 0) {
        alert("Cart is empty.");
        return;
    }

    const expensiveItems = cart.filter(item => item.price > 20000);

    if (expensiveItems.length === 0) {
        alert("No expensive items (>₹20,000) found in the cart.");
        return;
    }

    let message = "Expensive Items (>₹20,000):\n\n";
    expensiveItems.forEach(item => {
        message += `  - ${item.product}: ₹${item.price.toFixed(2)} (Qty: ${item.quantity})\n`;
    });

    alert(message);
}

```

7. Output:

alt text

8. Result/Conclusion:

The experiment successfully demonstrated the use of JavaScript arrays and objects for managing shopping cart data. The program calculates item totals, applies discount logic, and computes GST to generate the final invoice. Array methods such as `push`, `find`, and `filter` were used to add, search, and manage cart items efficiently. The case study also provided practical understanding of dynamic DOM manipulation and user interaction in JavaScript. Thus, the application successfully implements a functional and user-friendly cart total calculator with discount logic.